

PLSC 309: Quantitative Political Analysis

Problem Set 2

Weeks 3-4: Probability, Estimation, and Hypothesis Testing

Due: Sep-23-2026, 11:59 p.m. ET

Student Name: _____

Instructions

This problem set focuses on conceptual understanding. No mathematical calculations are required.

- Questions 1-20 are worth **0.2 point each**.
- Questions 21-30 are worth **0.3 point each**.
- The regular portion is worth **7 points**.
- The bonus question contains two parts worth **0.5 point each**, for up to **1 additional point**.
- For multiple-choice questions, select the single best answer.
- For short-answer questions, respond to every part. Brief and direct answers are sufficient.

Part I: Multiple Choice

20 questions x 0.2 point = 4 points

1. Random Variables

[0.2 points]

Which statement best describes a random variable?

- A. ☐ It assigns a numerical value to each possible outcome of a random process.
- B. ☐ It is any variable collected through random sampling.
- C. ☐ It must have an equal probability of taking every possible value.
- D. ☐ It is a value selected by the researcher before data collection.

2. Probability Distributions

[0.2 points]

What information does a probability distribution provide?

- A. ☐ The possible values of a random variable and their associated probabilities
- B. ☐ The names of the individuals included in a sample
- C. ☐ The difference between a population and a sample
- D. ☐ The probability that a research hypothesis is correct

3. Cumulative Distribution Functions

[0.2 points]

For a random variable X , what does the cumulative distribution function $F(x)$ represent?

- A. ☐ The probability that X is exactly equal to x
- B. ☐ The probability that X is greater than x
- C. ☐ The probability that X is less than or equal to x
- D. ☐ The expected value of X

4. Expected Value**[0.2 points]**

Which interpretation of the expected value of a random variable is most appropriate?

- A. ☐ The value that must occur in every sample
- B. ☐ The long-run average across many repetitions of a random process
- C. ☐ The largest possible value of the random variable
- D. ☐ The value with the greatest sampling variance

5. Central Tendency**[0.2 points]**

Which statement about expected value is correct?

- A. ☐ It describes one aspect of the center of a distribution.
- B. ☐ It completely describes the shape of a distribution.
- C. ☐ It must be one of the possible observed values.
- D. ☐ It measures the association between two variables.

6. Variance**[0.2 points]**

What does the variance of a random variable describe?

- A. ☐ The center of its distribution
- B. ☐ The dispersion of its values around the expected value
- C. ☐ The direction of association between two variables
- D. ☐ The number of observations in the population

7. Standard Deviation**[0.2 points]**

Why is standard deviation often easier to interpret than variance?

- A. ☐ Standard deviation is expressed in the same units as the original variable.
- B. ☐ Standard deviation is always smaller than the expected value.
- C. ☐ Standard deviation is unaffected by extreme observations.
- D. ☐ Standard deviation always falls between zero and one.

8. Covariance**[0.2 points]**

What does a positive covariance between two variables generally indicate?

- A. ☐ Larger values of one variable tend to be associated with larger values of the other.
- B. ☐ Larger values of one variable tend to be associated with smaller values of the other.
- C. ☐ The two variables have the same expected value.
- D. ☐ The two variables have the same variance.

9. Covariance and Measurement Units**[0.2 points]**

Which statement about covariance is correct?

- A. ☐ Its magnitude may change when the measurement units of a variable change.
- B. ☐ It must always fall between -1 and 1 .
- C. ☐ It is unaffected by the scales used to measure the variables.
- D. ☐ It measures only the dispersion of a single variable.

10. Correlation**[0.2 points]**

Which feature distinguishes correlation from covariance?

- A. ☐ Correlation is standardized and falls between -1 and 1 .
- B. ☐ Correlation can only be positive.
- C. ☐ Correlation is measured in the original units of both variables.
- D. ☐ Correlation measures the center of a distribution.

11. Population and Sample**[0.2 points]**

A researcher wants to study the political attitudes of all registered voters in a state but interviews 1,000 registered voters. What is the population?

- A. ☐ The 1,000 interviewed voters
- B. ☐ All registered voters in the state
- C. ☐ Only the voters who answered every survey question
- D. ☐ The set of recorded survey responses

12. Random Sampling**[0.2 points]**

Which procedure best represents a simple random sample?

- A. ☐ Surveying the first 100 people who respond to an online advertisement
- B. ☐ Surveying people outside one conveniently located polling place
- C. ☐ Selecting voters so that every voter has an equal chance of being included
- D. ☐ Asking current participants to recruit their friends

13. Sampling Variability**[0.2 points]**

Two researchers independently draw random samples from the same population and obtain slightly different estimates. Which statement is most appropriate?

- A. ☐ At least one researcher must have made a mistake.
- B. ☐ Different random samples can produce different estimates because of sampling variability.
- C. ☐ A random sample must always produce the true population value.
- D. ☐ The population parameter must have changed between the two studies.

14. Estimands**[0.2 points]**

Which statement best defines an estimand?

- A. ☐ The population quantity that the researcher wants to learn about
- B. ☐ The procedure used to select individuals for the sample
- C. ☐ The numerical result obtained from one particular sample
- D. ☐ The probability of rejecting the null hypothesis

15. Estimators and Estimates**[0.2 points]**

Which statement correctly distinguishes an estimator from an estimate?

- A. ☐ An estimator is a population parameter, while an estimate is a sample.
- B. ☐ An estimator is a rule or procedure, while an estimate is the numerical result obtained from a particular sample.
- C. ☐ An estimator is always unbiased, while an estimate is always biased.
- D. ☐ An estimator describes a population, while an estimate describes a probability distribution.

16. Unbiasedness and Efficiency**[0.2 points]**

Suppose Estimators A and B are both unbiased for the same population parameter, but Estimator A has a smaller sampling variance. Which statement is correct?

- A. ☐ Estimator A is more efficient than Estimator B.
- B. ☐ Estimator B is more efficient than Estimator A.
- C. ☐ Estimator A must produce the correct value in every sample.
- D. ☐ Estimator B must be biased.

17. Consistency**[0.2 points]**

An estimator is consistent when:

- A. ☐ It produces exactly the same estimate in every sample.
- B. ☐ It becomes increasingly concentrated near the true population parameter as the sample size grows.
- C. ☐ It has zero sampling variance in small samples.
- D. ☐ Its value becomes larger whenever the sample size increases.

18. Null and Alternative Hypotheses**[0.2 points]**

A researcher studies whether a new voter-mobilization program increases turnout. Which pair of hypotheses is most appropriate?

- A. ☐ H_0 : The program does not increase turnout; H_1 : The program increases turnout.
- B. ☐ H_0 : The program increases turnout; H_1 : The program does not increase turnout.
- C. ☐ H_0 : The sample is random; H_1 : The sample is not random.
- D. ☐ H_0 : The estimate is unbiased; H_1 : The estimate is efficient.

19. Test Statistics and Rejection Regions**[0.2 points]**

Which statement correctly describes a test statistic and a rejection region?

- A. ☐ The test statistic summarizes the evidence relative to the null hypothesis, and the rejection region contains values that lead to rejecting the null hypothesis.
- B. ☐ The test statistic gives the probability that the alternative hypothesis is true, and the rejection region contains values that confirm the null hypothesis.
- C. ☐ The test statistic guarantees that the sample is representative, and the rejection region identifies which observations should be removed.
- D. ☐ The test statistic eliminates Type I errors, and the rejection region eliminates Type II errors.

20. P Values**[0.2 points]**

Which statement best describes a p value?

- A. ☐ The probability that the null hypothesis is true
- B. ☐ The probability that the alternative hypothesis is true
- C. ☐ Assuming the null hypothesis is true, the probability of obtaining a result at least as extreme as the observed result
- D. ☐ The probability that the researcher has made a Type II error

Part II: Conceptual and Applied Questions**10 questions x 0.3 point = 3 points****21. Random Variables and Probability Distributions****[0.3 points]**

Let X represent the number of political campaign events attended by a randomly selected voter during the past year.

1. Explain why X is a random variable.
2. In this context, what information would the probability distribution of X provide?
3. What would the expected value $E(X)$ represent?

22. Center and Dispersion**[0.3 points]**

Two random variables have the same expected value, but Variable A has a larger variance than Variable B.

1. What is similar about the two distributions?
2. What is different about them?
3. Why is expected value alone insufficient to describe a distribution?

23. Covariance and Correlation**[0.3 points]**

A researcher finds that education and political knowledge have a positive covariance and a positive correlation.

1. What does the positive sign indicate about the direction of their association?
2. Suppose education is changed from years to months. Which measure - covariance or correlation - is more directly affected by this change in measurement units?
3. Why is correlation more convenient when comparing associations measured on different scales?

24. Evaluating a Sample**[0.3 points]**

A political campaign posts a voluntary survey on its social-media page and uses the responses to make claims about all voters in the state.

1. What is the intended population?
2. Why might the respondents not constitute a random sample?
3. What problem could this create for the campaign's conclusions?

25. Estimand, Estimator, and Estimate**[0.3 points]**

A researcher wants to know the proportion of all registered voters who support a proposed law. The researcher uses the proportion supporting the law in a random sample of 800 voters. In this sample, 56% support the law.

Identify:

1. The **estimand**
2. The **estimator**
3. The **estimate**

26. Comparing Estimators**[0.3 points]**

Estimators A and B are both unbiased estimators of the same population parameter. However, the sampling distribution of Estimator A is more tightly concentrated around the true parameter.

1. Which estimator is more efficient? Why?
2. Does unbiasedness require the two estimators to produce the same estimate in every sample? Explain briefly.

27. Applying the Central Limit Theorem**[0.3 points]**

Individual income is strongly right-skewed. A researcher takes many large, independent random samples and calculates the sample mean income for each sample.

1. What shape would you expect the distribution of the sample means to have?
2. Why can this occur even though individual incomes are not normally distributed?
3. Name the statistical principle that supports your answer.

28. Interpreting a Confidence Interval**[0.3 points]**

A statewide poll estimates that 52% of registered voters support a proposed policy. The reported 95% confidence interval is

[48%, 56%].

1. Interpret this 95% confidence interval from a repeated-sampling perspective.
2. Does the interval provide clear evidence that more than half of all registered voters support the policy? Explain using the values in the interval.
3. Is it correct to say that 95% of individual voters have opinions that fall between 48% and 56%? Why or why not?

29. Type I and Type II Errors**[0.3 points]**

A researcher tests the following hypotheses about a voter-mobilization program:

H_0 : The program does not increase voter turnout

H_1 : The program increases voter turnout

1. Describe a **Type I error** in the context of this study.
2. Describe a **Type II error** in the context of this study.
3. Which type of error occurs if the researcher concludes that the program increases turnout when it actually does not?

30. Interpreting a Hypothesis Test**[0.3 points]**

A researcher evaluates whether a voter-mobilization program increases turnout. The hypotheses are

H_0 : The program does not increase turnout

H_1 : The program increases turnout.

The researcher obtains a p value of 0.032 and uses a significance level of 0.05.

1. Should the researcher reject or fail to reject the null hypothesis?
2. Interpret the p value of 0.032 in the context of this study.
3. Does the p value mean that there is a 3.2% probability that the null hypothesis is true? Explain briefly.

Bonus Question

Two parts x 0.5 point = up to 1 bonus point

31A. Sample Size and Precision

[0.5 points]

Two researchers use the same random-sampling procedure to estimate statewide approval for the governor. Researcher A surveys 150 voters, while Researcher B surveys 1,500 voters.

Without performing calculations:

1. Which researcher's estimate would generally be more stable across repeated random samples?
2. Which researcher would generally obtain a narrower confidence interval?
3. Explain why.

31B. Failure to Reject the Null Hypothesis

[0.5 points]

A researcher conducts a hypothesis test using a small sample and obtains a p value of 0.18. The researcher states, "I have proven that the program has no effect."

1. Explain why this conclusion is not justified.
2. What does failure to reject the null hypothesis actually mean?
3. Explain how a Type II error could still have occurred.